

The free-roaming dog as a sentinel of human leptospirosis after the coastal El Niño event: an inference from the Peruvian evidence

Systematic review with meta-analysis of canine *Leptospira* spp. seroprevalence in Peru (2003–2023) and transportability analysis of the effect of the 2017 coastal El Niño

Jessica Ysabel Camacho García, Veterinarian and Zootechnician, CMVP 12434

DOI: <https://doi.org/10.5281/zenodo.23023822>

ORCID 0009-0002-6837-5311 · Zoovet Travel, Peru

Abstract

Background. The 2017 coastal El Niño event caused prolonged flooding, accumulation of solid waste and outbreaks of human leptospirosis along the northern coast of Peru. Free-roaming dogs are exposed to this environment earlier than the human population; however, no Peruvian studies are available in this population. The objectives of this study were to demonstrate, on the basis of the published Peruvian evidence, that *Leptospira* seroprevalence in unvaccinated dogs increases during El Niño events as a consequence of the environmental conditions they create; to assess the transportability of the effect observed in Lima to other settings; and to characterise the role of the free-roaming dog as a sentinel of human exposure within a One Health framework.

Methods. Peruvian literature published between 2003 and 2023 in indexed journals and thesis repositories was reviewed. Raw counts were extracted and a meta-analysis of proportions was performed using the logit transformation, a random-effects model (REML), the Hartung-Knapp confidence interval and a prediction interval, with pre-specified subgroup and sensitivity analyses. The transportability of the 2017 prevalence ratio published by Serrano-Martínez et al. (Lima) was assessed by conversion to an odds ratio and propagation of uncertainty through Monte Carlo simulation. All calculations were reproduced in two independent computing environments with hash verification.

Results. Twelve studies were identified, 10 with extractable data and 8 eligible for the baseline ($n = 993$ owned dogs). The pooled seroprevalence was 60.1% (Hartung-Knapp 95% CI 39.9–77.4), with $I^2 = 93.9\%$ and a prediction interval of 11.8–94.4%; sequential exclusion of studies shifted the estimate between 54.3% and 65.6%. The pooled estimate was 37.3% for peer-reviewed publications ($k = 3$) and 70.9% for undergraduate theses ($k = 5$). No study included free-roaming dogs or rodents from flooded areas or markets. The prevalence ratio of 3.11 (95% CI 1.61–6.03) for the event year in Lima was not applicable as a point multiplier, since 6.73% of plausible combinations produced prevalences greater than one; converted to an odds ratio and propagated with its uncertainty, the effect was positive across the entire range (2.5th percentile = 1.81; median 5.14). When the odds ratio is applied to the predictive distribution of the baseline, a coastal El Niño year would be associated with a seroprevalence in free-roaming dogs in an endemic coastal area of at least 36.4% in the conservative scenario and 89.4% in the central scenario. In coastal dogs the predominant serovar was Varillal (*L. licerasiae*); *Icterohaemorrhagiae* was documented in Lima and Chancay.

Conclusions. The available Peruvian evidence does not allow a national seroprevalence to be estimated, but it supports a consistent inference: the El Niño event increases *Leptospira* seropositivity in unvaccinated dogs in every plausible scenario, and after an event a substantial proportion of free-roaming dogs in an endemic coastal area would be seropositive: at least one in three, and nine in ten in the central scenario. This population, free of vaccine interference, is the only group in which a titre against serogroup *Icterohaemorrhagiae* can be interpreted as environmental exposure, and it has not been studied in the

country. A course of action is proposed for the clinician, based on sampling free-roaming dogs in flooded areas, interpreting the MAT by serogroup, keeping the positive animal alive for follow-up, and reporting the finding promptly. As a derived implication, the demonstrated increase in seroprevalence among unvaccinated dogs during the event supports the recommendation to vaccinate unimmunised domestic dogs before the onset of the phenomenon.

Keywords: leptospirosis; dogs; sentinel animal; El Niño; One Health; Peru; meta-analysis; *Icterohaemorrhagiae*.

1. Introduction

Leptospirosis is the most widely distributed bacterial zoonosis in the world, with a burden concentrated in tropical regions that combine heavy rainfall, inadequate sanitation and peridomestic rodent populations [1]. In Peru, the departments with the highest numbers of confirmed human cases have historically been Loreto, Cusco, Madre de Dios and Lima [2]. El Niño events and coastal cyclones shift this distribution towards the northern coast. The 2017 coastal event caused river overflows, mudflows and prolonged flooding in La Libertad, Piura and Lambayeque. In 2023, Cyclone Yaku reproduced this scenario in La Libertad: of 70 human leptospirosis cases with reactive serology, 65 were recorded in March, coinciding with the highest rainfall [3].

The environmental consequences of flooding are well characterised: accumulation of solid waste, persistence of standing water for weeks, and growth of the *Rattus norvegicus* population. This species is a maintenance host of serogroup *Icterohaemorrhagiae*, which it carries asymptomatically in the renal tubules and sheds persistently in its urine [1]. This serogroup causes the icteric form of the disease in dogs and Weil's disease in humans, and it has been consistently described in post-flood outbreaks in different regions of the world [1, 4].

In this setting, the free-roaming dog is the first mammal to come into prolonged contact with contaminated water, weeks before human cases appear. With respect to *Icterohaemorrhagiae*, the dog behaves as an accidental host: it neither maintains the serogroup nor transmits it efficiently to other dogs. Consequently, an antibody titre against *Icterohaemorrhagiae* in an unvaccinated dog is an indicator of environmental exposure rather than an individual finding, which corresponds to the operational definition of a sentinel animal and to the role that the One Health approach assigns to veterinary diagnosis in anticipating human disease.

Peru has a series of studies on *Leptospira* seroprevalence in dogs, systematised in a recent scoping review [5], and a single study that stratifies canine seropositivity by year and includes the summer of the 2017 coastal event [6]. Both sources lend themselves to two frequent and questionable uses: pooling the former into a national figure, and applying the latter as a multiplier to populations other than the original one. The present study set out to demonstrate, with an explicit method and on the basis of that evidence, that during an El Niño event *Leptospira* seroprevalence in unvaccinated dogs increases as a consequence of the associated environmental conditions (standing water, waste, rodents); to assess to what extent this effect is transportable; to document which animal populations have not been studied; and to propose a defined course of action for the veterinary clinician faced with a seropositive free-roaming dog after flooding. This demonstration also yields an implication for owners of unvaccinated domestic dogs, which is set out in the conclusions.

2. Methods

2.1 Design

A review of the published Peruvian evidence on *Leptospira* spp. seroprevalence in dogs was conducted, with a meta-analysis of proportions and a transportability analysis of a published effect. No primary data were collected and no animals were handled. The analysis plan (transformation, estimators, subgroups and sensitivity analyses) was fixed in writing on 25 September 2026, before extraction of the pending theses, and is recorded in the project data protocol (Appendix A).

2.2 Search and selection

The scoping review by Santana et al. (2025) [5] was used as an index, supplemented by searches in SciELO Peru, the RENATI and ALICIA repositories (CONCYTEC), Cybertesis UNMSM and PubMed Central. The inclusion criteria for the baseline were: cross-sectional design in dogs; prevalence estimable from raw counts (positives and total); diagnosis by the microscopic agglutination test (MAT); and a canine population not selected by clinical status or by human serological status. Studies with outcome-dependent sampling (clinically suspect dogs referred to a laboratory [7] and animals from households selected according to human serostatus [8]) were included only in sensitivity analyses. Case-control designs were excluded.

Twelve studies were identified, 10 of which allowed counts to be extracted. Two theses (Baldoceda 2019; Pacheco 2015) and one article (Salmón-Mulanovich 2019, open-access version) were reviewed in full text; one thesis (García 2016) was reviewed from the repository abstract only; one thesis (Romero 2014) was located, but its full text requires institutional authentication and was not extracted. The resulting count (12 identified, 10 extracted, 8 in the baseline, 2 in sensitivity analyses, 1 excluded, 1 not extracted) is obtained directly from the extraction file.

2.3 Extraction and verification

For each study, the following were recorded: the number of positives and the total as raw counts; the prevalence reported by the source, even when it differed from the quotient; the MAT cut-off; the serovar panel; the population; the locality and macro-region; the years of publication and sampling; and the type of publication (peer-reviewed or thesis). Discrepancies between the count and the published figure were retained and documented alongside the data point. In Céspedes 2004, 181/374 corresponds to 48.4% against a reported 52.2%; in Rondón 2018, 82.3% of 55 does not correspond to an integer, so 45/55 was used.

Undergraduate theses were treated with explicit reservations, verified in the full text where possible and recorded alongside the data point: inclusion of dogs with lapsed vaccination tested at a 1:100 cut-off, concentration of reactions at the lowest dilution, and sampling imbalances (73 males out of 80 animals, in one case). The internal arithmetic consistency of each thesis reviewed was checked table by table.

2.4 Meta-analysis of seroprevalence

Each proportion was transformed to the logit scale, with variance $1/x + 1/(n - x)$ and a continuity correction of 0.5 only when $x = 0$ or $x = n$. A random-effects model was fitted with τ^2 estimated by restricted maximum likelihood (REML); the DerSimonian-Laird estimator is reported as a sensitivity analysis. The confidence interval of the pooled effect was calculated with the Hartung-Knapp method (t distribution with $k - 1$ degrees of freedom and rescaled variance); the interval based on the normal

distribution is also reported to make the difference explicit when k is small. The 95% prediction interval was calculated according to Riley et al. (2011), with t on $k - 2$ degrees of freedom, and heterogeneity was quantified with Q , I^2 and τ^2 . The influence of each study was assessed by sequential exclusion.

The pre-specified subgroups were: macro-region (coast/rainforest), type of publication, MAT cut-off, population, and sampling period (before 2015 / 2015 onwards). A subgroup was analysed only when it had at least two levels with at least two studies each; otherwise it was declared non-analysable. Studies without a sampling year did not constitute a level. Differences between subgroups were tested with the between-group Q statistic under a common τ^2 . Sensitivity analyses comprised the individual and joint inclusion of the studies classified in that category and the exclusion of studies with $n < 10$. Neither a funnel plot nor Egger's test was performed, since $k < 10$.

2.5 Transportability of the El Niño effect

Serrano-Martínez et al. [6] report, for 271 clinically suspect dogs in Lima, the prevalence by year (2014 to 2017) and the prevalence ratio (PR) adjusted by Poisson regression with robust standard errors, with 2014 as the reference category. Cell counts were taken from Tables 1, 2 and 3 of the published version, and each cell was verified to reproduce the published PR. One printed cell (12 positives for 2015) proved incompatible with the rest of its table: the printed percentages, the total stated in Table 1 of that article, the column totals and the PR itself all require 19 positives. The figure is cited as published and the calculation uses 19; the four internal checks are detailed in Appendix B.

A prevalence ratio multiplies a baseline prevalence; with $p_{2014} = 16.7\%$, any PR greater than 6.00 yields a prevalence greater than one. To assess the transportability of the effect, the PR was converted to an odds ratio in the source population and uncertainty was propagated by Monte Carlo simulation (200,000 iterations; PR with a lognormal distribution fitted to the published interval; p_{2014} with a Jeffreys Beta distribution). The fraction of arithmetically impossible iterations and the distribution of the odds ratio conditional on the admissible region are reported. As a complementary analysis, the odds ratio was applied to the predictive distribution of the meta-analysis (and not to its confidence interval, since the target population does not correspond to any of the pooled studies) in order to obtain seroprevalence scenarios after an event.

2.6 Reproducibility

The two calculation modules were implemented as Python programs that record the seed, the input file and its SHA-256 digest, and halt execution if the digest does not match the declared value. Text outputs are generated with normalised line endings and paths, so that they are byte-for-byte comparable across operating systems. Each module was run in two independent environments (Windows 11, Python 3.13, NumPy 2.3.3; Linux, Python 3.11, NumPy 2.4.4) with identical digests for all text outputs, and was audited by independent reimplementations of the calculation. The digests are listed in Appendix A.

3. Results

3.1 Included studies

Table 1 summarises the ten studies with extractable data. The eight baseline studies total 993 owned dogs, from rural communities in Ucayali, Chancay, Piura, Amazonas and Madre de Dios, from a buffer zone in Lambayeque and from a private clinic in Lima. The known sampling years range from 1999 to

2019; two studies do not report them. Seven studies used a 1:100 cut-off and one does not report it. The number of serovars in the panel ranged from 5 to 27.

Study	Locality	Population	Sampling	Cut-off	Pos/n	%	Type
Céspedes 2004 [9]	Coronel Portillo, Ucayali	owned	2001	1:100	181/374	48.4	journal
Céspedes 2007 [10]	Chancay, Lima	owned	2001	1:100	67/241	27.8	journal
Sacsquispe 2003 [11]	Salitral and Malacasí, Piura	owned	1999	1:100	1/3	33.3	journal
Pacheco 2015 [12]	J. I. Távara Pasapera, Piura	owned	2009–2012	1:100	45/80	56.3	thesis
García 2016 [13]	Infierno Native Community, Madre de Dios	owned	NR	1:100 (5 serovars)	17/30	56.7	thesis
Rondón 2018 [14]	Corosha, Amazonas	owned	2017	NR	45/55	81.8	thesis
Baldoceda 2019 [15]	S. J. de Miraflores, Lima (clinic)	owned	2019*	1:100	93/106	87.7	thesis
Atalaya 2023 [16]	Sta. Rosa de las Salinas, Lambayeque	owned	NR	1:100	65/104	62.5	thesis
Siuce 2015 [7] — sensitivity	Metropolitan Lima	clinical suspects	2012–2014	1:100	177/305	58.0	journal
Salmón 2019 [8] — sensitivity	Tumbes, Lima, Cusco, Madre de Dios	households by human serostatus	2014	1:100	44/53	83.0	journal

Table 1. Studies with extractable data. NR: not reported. *Year inferred from the text (“summer of the present year”; thesis defence record dated November 2019). Excluded: Huerta 2013 [17], case-control design. Located but not extracted: Romero 2014 [18], Nuevo Sullana, Piura, n = 91.

3.2 Pooled effect and heterogeneity

The pooled seroprevalence of the eight baseline studies was 60.1% (Hartung-Knapp 95% CI 39.9–77.4; normal CI 43.0–75.0), with $\tau^2 = 0.857$, $I^2 = 93.9\%$ and $Q = 115.6$ (7 df; $p < 0.001$). The 95% prediction interval was 11.8–94.4%. With the DerSimonian-Laird estimator the estimate was identical (60.1%; HK CI 39.9–77.4). The crude prevalence of the summed counts was 51.8%; the difference from the pooled estimate reflects the flattening of random-effects weights when τ^2 is high: the three-animal study received 5.3% of the weight and the 374-animal study 14.4% (Figure 1).

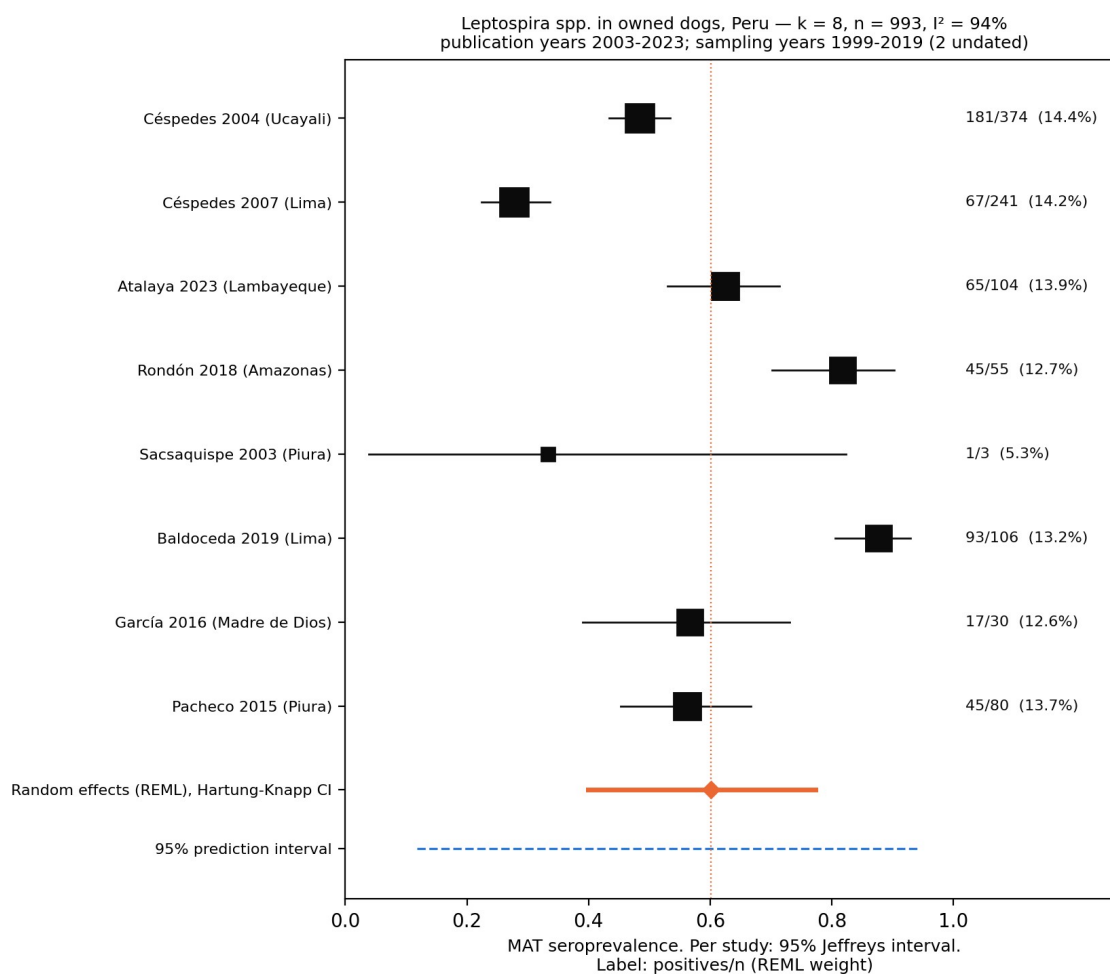


Figure 1. MAT seroprevalence of *Leptospira* spp. in owned dogs, Peru. Per study, 95% Jeffreys interval; label: positives/total (REML weight). Diamond: pooled effect with Hartung-Knapp CI. Dashed line: 95% prediction interval.

Sequential exclusion of each study shifted the pooled estimate between 54.3% (without Baldoceda 2019) and 65.6% (without Céspedes 2007), with I² between 89.6% and 94.8% (Table 2). No single study accounted for the observed heterogeneity.

Excluded study	k	Pooled %	95% CI (HK)	95% PI	I ² %
Céspedes 2004	7	61.9	38.2–81.1	9.4–96.2	94.8
Céspedes 2007	7	65.6	46.7–80.5	19.0–93.9	89.6
Atalaya 2023	7	59.6	35.5–79.9	8.1–96.1	94.5
Rondón 2018	7	56.2	34.9–75.4	9.7–93.9	93.8
Sacsaquispe 2003	7	61.6	39.4–79.8	10.6–95.6	94.8
Baldoceda 2019	7	54.3	36.4–71.2	13.4–90.2	91.3
García 2016	7	60.5	36.5–80.3	8.6–96.1	94.8
Pacheco 2015	7	60.6	36.5–80.5	8.5–96.2	94.7

Table 2. Influence analysis by sequential exclusion, REML with Hartung-Knapp CI.

3.3 Subgroups and sensitivity analyses

Two of the five pre-specified subgroups were not analysable with $k = 8$: the cut-off (seven studies at 1:100 and one not reported) and the population, since all eight studies concern owned dogs. The results of the analysable subgroups and of the sensitivity analyses are presented in Table 3.

Subgroup	Level	k	n	Pooled %	95% CI (HK)	Q between groups (p)
Macro-region	coast	5	534	57.6	24.3–85.2	0.10 (0.756)
	rainforest	3	459	63.1	18.0–93.0	
Type of publication	peer-reviewed	3	618	37.3	13.9–68.8	3.67 (0.055)
	undergraduate thesis	5	375	70.9	47.9–86.5	
Sampling period	before 2015	4	698	42.7	22.8–65.3	5.65 (0.017)
	2015 onwards	2	161	85.5	24.3–99.1	
Sensitivity	+ Since 2015	9	1298	59.8	42.6–75.0	—
	+ Salmón 2019	9	1046	63.2	44.4–78.7	—
	+ both	10	1351	62.6	46.2–76.6	—
	$n \geq 10$ only	7	990	61.6	39.4–79.8	—

Table 3. Pre-specified subgroups and sensitivity analyses. Two studies without a sampling year were not included in the period subgroup.

Two results require cautious interpretation. The difference between peer-reviewed publications and theses (37.3% versus 70.9%) did not reach conventional significance, but its magnitude is considerable and it has a material influence on the baseline estimate. The difference by sampling period (42.7% before 2015 versus 85.5% from 2015 onwards; $p = 0.017$) cannot be read as a temporal trend: the two studies sampled from 2015 onwards are Rondón 2018 and Baldoceda 2019, the two theses with the highest seroprevalence, whereas three of the four earlier studies are peer-reviewed publications. With $k = 8$, the partitions by period and by type of publication practically coincide, and the data do not allow the two factors to be separated.

No sensitivity analysis altered the interpretation: inclusion of the two studies with non-population-based sampling placed the pooled estimate between 59.8% and 63.2%, and exclusion of the three-animal study placed it at 61.6%.

3.4 Serogroup distribution in the studies reviewed in full text

In Piura, Varillal (*L. licerasiae*, serogroup Iquitos) was identified in 38 of the 45 seropositive dogs (84.4%), with titres of up to 1:1600; Canicola in 5; and Icterohaemorrhagiae in none [12]. In Lima, of 133 reactions in 93 positive dogs, Iquitos accounted for 54 (40.6%), Icterohaemorrhagiae for 36 (27.1%) and Canicola for 5 (3.8%) [15]. In Chancay, Céspedes 2007 reported Icterohaemorrhagiae and Canicola as the most frequent serogroups [10]. The only study with a five-serovar panel, lacking Varillal and Iquitos [13], was unable to detect the predominant coastal serogroup, which suggests that its estimate of 56.7% underestimates exposure.

3.5 Transportability of the El Niño effect

The cells from Serrano-Martínez et al. [6] exactly reproduced the published prevalence ratios (Table 4). With $p_{2014} = 9/54 = 16.7\%$ and $p_{2017} = 18/26 = 69.2\%$, the crude PR was 4.15 and the crude odds ratio 11.25. The published adjusted PR of 3.11 implies $p_{2017} = 51.8\%$ and an odds ratio of 5.38. With this baseline prevalence, any PR greater than 6.00 yields a prevalence greater than one, and the published upper limit of the PR is 6.03; the interval therefore exceeds the admissible range. This feature is not an error on the part of the authors (Poisson regression with robust standard errors does not restrict the result to the interval $[0, 1]$, unlike the log-binomial model), but it means that the estimate cannot be applied directly to populations with a different baseline prevalence.

Stratum	Positive	n	%	Bivariate PR	Adjusted PR (95% CI)
2014 (reference)	9	54	16.7	ref.	ref.
2015	19 (printed: 12)	90	21.1	1.27	1.25 (0.61–2.58)
2016	47	101	46.5	2.79	2.62 (1.39–4.29)
2017 (coastal El Niño)	18	26	69.2	4.15	3.11 (1.61–6.03)
Spring–summer	59	132	44.7	1.83	1.82 (1.28–2.59)
Autumn–winter (ref.)	34	139	24.5	ref.	ref.

Table 4. Cells from Serrano-Martínez et al. 2020 [6], Tables 1–3 of the original article. For 2015 the printed figure is cited and the calculation uses 19; see Appendix B.

The simulation (Figure 2) quantified this limitation: 6.73% of iterations produced prevalences greater than one, and the transportable odds ratio, conditional on the admissible region, had a median of 5.14 with a 95% CI of 1.81–53.80 (25th, 75th, 90th and 95th percentiles: 3.40; 8.70; 16.95; 29.63). The ratio of 29.8 between the limits of the interval rules out its use as a point multiplier. The result was insensitive to the order in which the random variables were generated (6.78% and 5.15 with the reverse order). For comparison, the odds ratio range of 1.8 to 4.2 for heavy rainfall reported in a recent systematic review [19] is a between-study range and not an estimate with an interval; the median obtained here lies outside it.

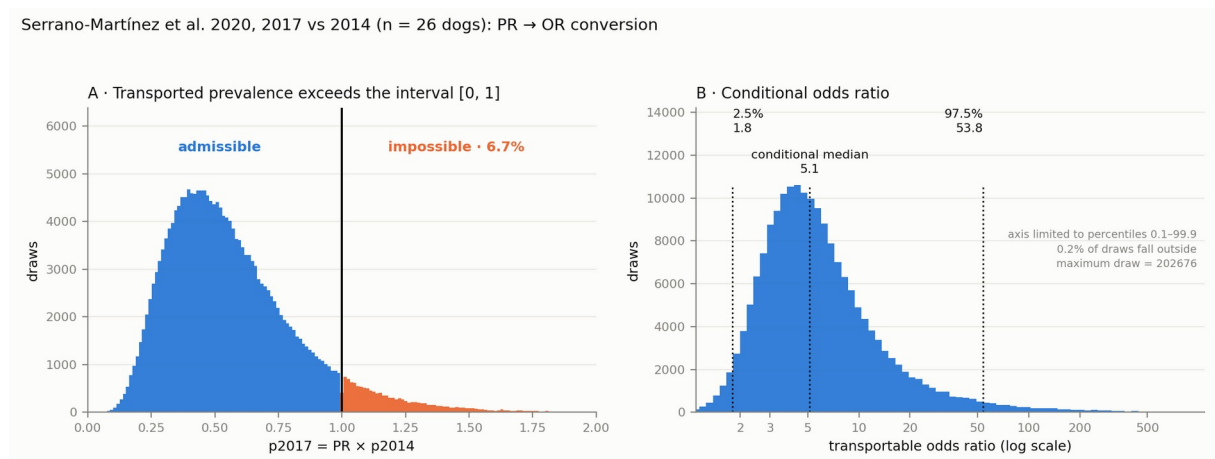


Figure 2. (A) Product $PR \times p_{2014}$ over 200,000 iterations; 6.7% exceed the interval $[0, 1]$. (B) Resulting odds ratio, conditional on the admissible region, on a logarithmic scale; the 95% CI spans 1.8 to 53.8.

Although it cannot be used as a point multiplier, the transportable odds ratio allows an inference by scenarios. Applied to the predictive distribution of the meta-analysis baseline at a locality not included in the studies (median 60.1%; prediction interval 11.9–94.4, consistent with that reported in 3.2), it yields,

for a year with a coastal El Niño event, a median seroprevalence of 89.4% with a 95% CI of 36.4–99.5 (Table 5). In the conservative scenario, corresponding to the lower limit of the interval, at least 36.4% of free-roaming dogs in an endemic coastal area would be seropositive after the event; in the central scenario, 89.4%. In no plausible scenario is the effect null: the 2.5th percentile of the odds ratio is 1.81 and the lower limit of the published PR is 1.61. The width of the interval, 63 percentage points, expresses the precision that a direct measurement in this population could provide; it does not affect the direction of the effect or its lower limit.

Table 5. Scenario-based inference for free-roaming dogs in an endemic coastal area (inferred values, not measured)

Scenario	Definition	Predictive baseline (no event)	After a coastal El Niño year	Approximate proportion
A — conservative	lower limit of the 95% PI	11.9%	36.4%	one in three
B — central	median	60.1%	89.4%	nine in ten
C — upper	upper limit of the 95% PI	94.4%	99.5%	virtually all

Baseline: predictive distribution of the meta-analysis ($k = 8$), sampled with t on $k - 2$ df, as for the prediction interval in 3.2. Effect: conditional odds ratio transported from Serrano-Martínez et al. 2020 [6], 200,000 iterations. The predictive baseline refers to owned dogs; in free-roaming dogs, with greater exposure and no vaccination, scenario A should be interpreted as a lower bound.

4. Discussion

4.1 Scope of the Peruvian evidence

The ten studies with available data do not allow the proportion of dogs in Peru with antibodies against *Leptospira* to be estimated. The limitation lies not in the scarcity of data but in the fact that they cannot be meaningfully pooled: a prediction interval of 11.8% to 94.4% and a pooled estimate whose leave-one-out values span eleven percentage points do not describe a population, but rather the distance between studies conducted in different localities, with different panels and different populations. The pooled figure is reported as evidence of this heterogeneity and not as a national estimate.

Two features of the evidence deserve attention. First, the pooled estimate is around 37% for the three peer-reviewed publications and around 71% for the five undergraduate theses. The theses reviewed in full text are internally consistent, but they incorporate methodological decisions that tend to raise seropositivity (inclusion of dogs with lapsed vaccination tested at a 1:100 cut-off, with 77 of 133 reactions at the lowest dilution [15]) or unexplained sampling imbalances [12]. These reservations are recorded alongside each data point so that the reader can weigh the provenance of each figure.

Second, the eight baseline studies concern owned dogs. No measurement of *Leptospira* was identified in the Peruvian literature in free-roaming dogs, in rodents from food markets, or in dogs or rodents from areas flooded after an El Niño event. The only approximation is the 1999 report from Salitral, Piura [11], with three dogs and twelve rodents, which constitutes a design precedent and not a prevalence estimate. This gap corresponds to the population that the present study identifies as the sentinel.

4.2 Inference from the El Niño effect observed in Lima

The prevalence ratio of 3.11 for the year of the coastal event [6] is the only Peruvian data point linking El Niño with canine seropositivity. It cannot be applied to other populations as a point multiplier, for the arithmetic reasons set out in the results; propagated with its uncertainty, however, it supports an inference by scenarios (Table 5). There is also a mechanistic consideration that calls for interpreting this

inference with caution and, at the same time, as a lower bound: the Lima authors exclude temperature and humidity from the model because of collinearity with season, so the effect they measure is thermal in nature. In La Libertad, both the 2017 coastal event and Cyclone Yaku in 2023 manifested as rainfall, river overflow and flooding; during Yaku, 65 of 70 human cases were concentrated in March, the month of highest rainfall [3]; the mechanism is hydrological. Transferring a coefficient from one mechanism to another is not an approximation; it conflates two distinct causal mechanisms. The year 2017 in Lima, with 26 clinically suspect dogs, also reflects the superimposition of a secular trend, changes in the sampled population and the event itself.

What the analysis does allow us to state is that the effect of El Niño on canine seropositivity is positive across the entire plausible range, and that after an event a substantial proportion of free-roaming dogs in an endemic coastal area would be seropositive, between one in three and nine in ten. What it does not allow is a single figure: to claim that El Niño triples canine leptospirosis in Peru would amount to selecting the median and omitting the interval. The interval, as reported, is sufficient to support the clinical course of action proposed.

4.3 The free-roaming dog as a sentinel of *Icterohaemorrhagiae*

With respect to *Icterohaemorrhagiae* the dog is an accidental host, so a titre in a free-roaming dog after flooding reflects environmental exposure (water, mud, rodent urine) and not a characteristic of the animal. The distinction from *Canicola* is essential: the dog is the maintenance host of *Canicola* and transmits it to other dogs, so a positive result for this serogroup provides information about the canine population and not about the flooding. *Canicola* therefore falls, by definition, outside the sentinel rationale.

The Peruvian data require a third serogroup to be considered. Varillal, a serovar of *Leptospira licerasiae*, predominates in coastal dogs: 38 of 45 seropositive animals in Piura [12] and 54 of 133 reactions in Lima [15]. This species, of intermediate pathogenicity, was isolated from febrile patients in Iquitos and from peridomestic *Rattus norvegicus* and *R. rattus* [20]; it has been detected in asymptomatic persons in Lima with a dog at home and rodents present [21]; and the brown rat carries it after experimental infection [22]. It causes a mild febrile illness and not the severe icteric form. Its relevance to the argument is twofold: it confirms that the rodent–water–dog chain operates on the Peruvian coast, and it explains why a panel without Varillal underestimates exposure. Nevertheless, the sentinel with implications for clinical practice is defined by *Icterohaemorrhagiae*, a serogroup that is virulent in both dogs and humans, associated with post-flood outbreaks and included in all the panels used.

The interpretability of this indicator depends on a condition that turns the evidence gap into an argument. Canine vaccines contain *Icterohaemorrhagiae* and *Canicola*; a vaccine titre at 1:100 is indistinguishable from an infection titre, which is why Serrano-Martínez et al. raised the cut-off to 1:400 in vaccinated dogs or dogs of unknown vaccination status [6]. The free-roaming dog is not vaccinated and therefore has no interfering vaccine titres. The population in which the MAT against *Icterohaemorrhagiae* can be interpreted unambiguously is, consequently, the one that has not been studied.

4.4 Course of action proposed to the veterinary clinician

The foregoing leads to a clinical course of action, not a research protocol:

1. **Sample in flooded areas.** The food market is one of the points of exposure, but not the only one: the flooded neighbourhood, the mudflow channel and the sites where waste accumulates, where

the rodent population is concentrated, are the places where the free-roaming dog acquires the infection.

2. **Interpret the MAT by serogroup.** In a free-roaming dog, a titre against Icterohaemorrhagiae after flooding constitutes epidemiological information about the human population of the area; a titre against Canicola provides information about the canine population; a titre against Varillal indicates contact with water contaminated by rodents.
3. **Keep the positive animal alive for follow-up.** A dead animal provides a single data point; a live, identified and traceable animal allows a series. The clinician's first questions (whether the animal is alive and whether it has had contact with people) determine the epidemiological usefulness of the result.
4. **Report the finding.** The veterinarian who identifies a free-roaming dog seropositive for Icterohaemorrhagiae after flooding holds a data point that the human health care system will not obtain until cases appear. The present study does not evaluate the performance of any institution; it describes the information and who should receive it.

4.5 Limitations

The analysis is based on eight studies, five of them undergraduate theses reviewed in full or in part, and includes no study from La Libertad. The only study stratified by year has 26 animals in the year of interest and a population of clinical suspects; one of its printed cells is arithmetically incompatible with the rest of the table, is cited as published, and the calculation uses the figure that the table itself requires. The Salitral precedent comprises three dogs. The predominant serogroups in Peruvian dogs do not coincide with the one proposed as the sentinel, a discrepancy that is addressed explicitly rather than omitted. The hydrological mechanism in La Libertad is inferred from the monthly distribution of human cases during Cyclone Yaku in 2023, and not from measurements in animals or from the 2017 event. The transportability analysis applies an adjusted PR to a crude prevalence; this is an approximation, the only one possible with the published information, and it is declared as such.

4.6 Unresolved questions

It remains to be determined where within the interval the seroprevalence of free-roaming dogs after flooding lies (closer to 36% or to 89%), how quickly it is reached, and the relative proportions of Icterohaemorrhagiae and Varillal. The present study bounds the range and establishes the direction of the effect; a direct measurement in this population would provide the precision. The proposed clinical course of action is the same at any point in the range.

5. Conclusions

- The published Peruvian evidence on *Leptospira* in dogs does not aggregate into a single national estimate, but it establishes a lower bound: the three peer-reviewed publications, the most conservative, place exposure above one in three owned dogs, and the pooled estimate at six in ten.
- The effect of the coastal El Niño event observed in Lima, propagated with its uncertainty, allows the inference that after an event a substantial proportion of free-roaming dogs in an endemic coastal area would be seropositive: at least one in three in the conservative scenario and nine in ten in the central scenario. No point multiplier can be derived; the resulting range is sufficient to support the proposed course of action.

- No Peruvian study of *Leptospira* in free-roaming dogs or in rodents from flooded areas or markets has been identified. These populations are both the most exposed and the only ones in which the MAT against *Icterohaemorrhagiae* can be interpreted without vaccine interference.
- The free-roaming dog seropositive for *Icterohaemorrhagiae* after flooding is a sentinel of human exposure: since it is not vaccinated, its titre reflects the environment. Varillal is an additional marker of environmental circulation on the coast. Canicola provides no information about the environment.
- For the veterinary clinician: sampling free-roaming dogs after flooding, in the affected area and not only in the market, has a high yield in any scenario; the MAT should be interpreted by serogroup; the positive animal should be kept alive for follow-up; and the finding should be reported to the appropriate authority.
- During an El Niño event, *Leptospira* seroprevalence in unvaccinated dogs increases as a consequence of the environmental conditions the event creates. Consequently, owners of unvaccinated domestic dogs should have them immunised before the onset of the phenomenon. This implication derives from the present study and constitutes the starting point of a subsequent study.

Derived implication

The results of this study (the increase in *Leptospira* seroprevalence in unvaccinated dogs during El Niño events in every plausible scenario, and the high proportion of free-roaming dogs that would be seropositive after an event in an endemic area) have a consequence that goes beyond the sentinel function. The unvaccinated domestic dog shares with the free-roaming dog the same environment, the same water and the same rodent population; the difference between them lies in the existence of an owner who can decide on its care. The recommendation to vaccinate unimmunised domestic dogs before the onset of the El Niño season derives directly from this evidence. It is not the central aim of the study but its natural consequence, and it defines the starting point of the next study.

Contributions and reproducibility

Conception, design, clinical interpretation and writing: Jessica Ysabel Camacho García. Data curation, source verification, implementation of the calculation modules and computational reproducibility: Carlos Eduardo Ravello Joo (data curator). The data files, the programs and the SHA-256 digests of all outputs are available in the public repository *leptospira-peru-modelado*, version 1.0 [23], and are described in Appendix A.

Conflicts of interest: none declared. Funding: none. Ethical considerations: the study is based exclusively on published data; no animals were handled and no primary data were collected.

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Appendix A. Reproducibility

Public repository: *leptospira-peru-modelado*, version 1.0, <https://doi.org/10.5281/zenodo.23024275> [23]. It contains the data files, the programs and the outputs whose digests are listed below.

The two calculation modules (*src/m1_base.py*, *src/m2_multiplicador.py*) are Python programs that read data files with a declared SHA-256 digest and halt execution if it does not match. Random seed: 20260925. They were run in two independent environments and the text outputs were identical byte for byte.

Environment	Python	NumPy	pandas	SciPy	statsmodels	matplotlib
Windows 11 (Miniconda)	3.13.5	2.3.3	2.3.3	1.18.1	0.15.0	3.10.7
Linux (container)	3.11.15	2.4.4	3.0.2	1.17.1	0.15.0	3.10.9

Table A1. Execution environments.

File	SHA-256 (first 16)
datos/extraccion_seroprevalencia_canina_peru.csv	cefd58f97640be1b
datos/serrano2020_celdas.csv	4add776474228163
salidas/m1_base_resultados.md	14fbce84d9dd880f
salidas/m1_base_agrupada.csv	00b9217cfe59c033
salidas/m1_base_influencia.csv	d6fd668bd4421ab9
salidas/m2_mult_resultados.md	ce19b5fb9be89f5d
salidas/m2_mult_montecarlo.csv	5abbbab72bf2da97

Table A2. Input and output hashes, identical in both environments. Each module was also audited by independent reimplementations of the calculation (REML by optimisation of the restricted likelihood against the fixed-point algorithm; Monte Carlo reconstructed from the printed parameters), with agreement to six decimal places.

Appendix B. The 2015 figure in Serrano-Martínez et al. 2020

Table 2 of the publication records 12 positives for 2015. Four internal consistency checks within the article indicate that the figure consistent with the rest of the data is 19:

1. The same table prints 21.1% and 78.9% with 71 negatives. If 71 is 78.9%, the total is 90, and 21.1% of 90 is 19.
2. Table 1 of the publication states $n = 90$ for 2015. With 12 positives and 71 negatives the total would be 83.
3. The column totals are 93 positives and 178 negatives out of 271 dogs. With 12, the positives by year add up to 86 and the dogs to 264; the rows by season do add up to 93 and 271.
4. The published bivariate PR for 2015 is 1.27, which results from $(19/90)/(9/54) = 1.267$. With 12/83 it would be 0.87.

The text cites the figure as published and the calculation uses the one that the table itself requires. The 2015 cell is involved only in the validation of the cells; the transportability analysis uses 2014 and 2017 and is not affected by it.